

Amirul Hakim

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EDUCATION

Bachelor of Engineering in Mechanical Engineering | Universitas Pancasila

September 2022 – July 2026

Cumulative GPA: 3.72/4.00

Undergraduate Thesis: Effect of Feed Rate in Micro-Milling on Burr Height and Dimensional Accuracy of Titanium I-Plate Implants; experimental machining of Commercially Pure Titanium with micro-milling method and analyzed optical dimensional and burr-height measurement using digital microscopy and image processing software ImageJ. The research showed that

EXPERIENCE

Manufacturing Engineering Intern | PT Defa Angkasa Utama

August 2025 – October 2025

- Analyzed end-to-end sheet-metal HVAC louver production and mapped 31 operation/inspection stages across an approximately 300-minute production cycle using an Operation Process Chart (OPC) and helped with standardizing process with Standard Operating Procedure (SOP), increasing work efficiency by 22%.
- Identified manual welding setup and component positioning as repeatability constraints and proposed a modular welding-jig concept for configurable louver sizes potentially saving welding process time up to 60%.

Maintenance and Instrumentation Intern | PT Timah Industri

September 2024 – February 2025

- Developed an Arduino/ESP32-based monitoring system for chiller system, acquiring field data through RS-485/Modbus RTU and publishing telemetry via MQTT into MySQL-backed dashboard for real-time and historical monitoring of flow, temperature, and related operating data.
- Assisted with design and implementation of a monitoring/control pH and Total Dissolved Solids (TDS) system based on Programmable Logic Controller for a Water Treatment Plant (WTP), integrated into centralized monitoring system, reducing potential total down time by 40%.
- Contributed to MEP planning and design with Autodesk AutoCAD and Revit for a production control-room expansion.

Mechanical Drafter | PT Jordyla Mitra Sejati

July 2023 – May 2024

- Produced and revised 2D manufacturing drawings and 3D CAD models for machined and fabricated components used in heavy-machinery and automotive applications.
- Prepared cutting profiles, part layouts, BOMs, drawing revisions, geometric dimensioning and tolerancing (GD&T), CNC programs, and other production documentation where applicable.
- Performed reverse engineering of undocumented components and coordinated drawing revisions with production personnel across machining, laser cutting, bending, welding, and assembly processes.
- Maintained drawing files, revisions, part numbers, and other technical documentation.

PROJECTS

PLC/HMI Industrial Pumping System

Developed a PLC-based pumping control system with automatic duty/standby sequencing, level control, motor interlocks, fault handling, alarms and HMI visualization. Implemented manual/automatic operating modes and simulated abnormal operating conditions to validate control logic and equipment protection.

Modular Roller Conveyor Mechanical Design

Designed a modular non-powered roller conveyor for manual material handling, incorporating straight and curved conveyor sections, structural frames, support legs, connectors, stoppers, and bearing-supported roller assemblies. Developed the complete CAD assembly, component design, BOM, and five-sheet production drawing package for fabrication and assembly.

CNC Energy Monitoring & Process Analytics System

Developed ESP32 firmware with task-separated sensor acquisition and storage, job-level energy accumulation, RTC timestamping, and SD-card CSV logging; served a local web dashboard with real-time, History, Detail, and Compare views for analyzing energy consumption across machining runs.

CERTIFICATIONS

Certified SOLIDWORKS Professional (CSWP) | Dassault Systèmes

September 2024

Certificate of Competency in 3D CAD Drafting | BNSP

July 2024 – July 2027

Siemens PLC & HMI Programming Specialization | Packt, Coursera

March 2026

SKILLS

CAD & Design: SolidWorks, Autodesk Inventor, CAD/CAM, 3D modeling, assembly design, 2D manufacturing drawings, GD&T.

Manufacturing: CNC machining, turning, milling, drilling, laser cutting, sheet-metal fabrication, press-brake bending, MIG welding, jig & fixture design, SOP, OPC.

Automation & Data: Siemens PLC, TIA Portal, HMI, Arduino/ESP32, Modbus RTU, RS-485, MQTT, MySQL, industrial sensor monitoring.

Language: Indonesian (Native), English (Professional Working Proficiency), German (Intermediate Level)